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AI Diabetic Retinopathy Screening in a Primary Care Setting in Rural Maine

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INTRODUCTION

Diabetic retinopathy (DR) is the leading cause of blindness globally. Early diagnosis and treatment reduce the risk of vision loss by 98%.¹ Standard of care for patients with diabetes includes annual DR screenings; the Centers for Medicare and Medicaid (CMS) DR screening target is 67%. Screening rates in the USA are approximately 60%.² To increase access, the FDA first approved artificial intelligence (AI) DR screening in 2018,³ which meets Healthcare Effectiveness Data and Information Standards for quality of care.⁴

A recent systematic review by Zhelev et al. concluded that AI had a higher sensitivity ($> 85\%$) versus human graders in the detection of referable diabetic retinopathy.⁵ Several studies have demonstrated a reduction in both health system and societal costs with AI.⁶ In Rwanda, an RCT showed AI's immediate feedback increased rates of specialty follow-up within 30 days versus human grading.¹ AI for DR screening may improve outcomes and address healthcare disparities through increasing access and efficiency while decreasing cost.⁷

Persistently low screening rates with poor follow-up at a rural clinic led this team to introduce AI DR screening technology. Our prior study found patients and providers had a positive experience with the tool.⁴ This follow-up study examines the tool's impact on screening and referral rates.

METHODS

This retrospective cohort study (January 2021 – June 2023) examined outcomes of adult (> 18 years) diabetic patients screened for DR at Western Maine Primary Care (WMPC) in Norway, ME, using the AI EyeArt EyeNuk technology. Positive DR screens received an urgent referral to an eye care center; negatives did not receive a referral. Prior to AI DR screening, all diabetics from both sites were referred to an eye care center annually. DR screening rates between December 2019 and June 2023 were tracked for quality metrics at both WMPC and Franklin Health Primary Care (FHPC), an equivalent rural practice that did not implement AI for DR screening. Summary statistics were calculated for the screened patients. Chi-square tests of independence and unadjusted odds ratios were calculated to compare the proportions of patients screened at WMPC and FHPC prior to AI technology implementation and the month of study completion. All analyses used R-Studio (version 4.2.1). The MaineHealth Institutional Review Board deemed the study exempt.

RESULTS

Three hundred twenty patients underwent EyeArt screening at WMPC. Of these, 19% ($n = 61$) screened positive for DR and were urgently referred to an optometrist/ophthalmologist (Table 1). Fifty-one percent of patients who followed up at an eye care center were found to have DR and received appropriate treatments. The follow-up rate among those who screened positive was 74% with an average time of 36 days. Before AI technology implementation in December 2020, patients seen at WMPC were 14% less likely to be screened for DR than at FHPC (OR = 0.86, 95% CI, 0.74, 0.99, $p = 0.041$). After study completion in June 2023, patients seen at WMPC were 2.8 times more likely to be screened for DR than those seen at FHPC (OR = 2.82, 95% CI, 2.42, 3.27, $p < 0.001$) (Fig. 1).

Table 1 Outcomes of Patients Who Screened Positive for DR Using EyeArt EyeNuk Technology at WMPC Between January 2021 and June 2023

Figure 1

Percentage of diabetic patients screened for diabetic retinopathy at Western Maine Primary Care (WMPC) and Franklin Health Primary Care (FHPC) between December 2019 and June 2023. The dashed vertical line indicates the implementation of the AI DR screening tool. The monthly percentages of patients screened for DR are represented by the solid green line (WMPC) and the solid orange line (FHPC). The solid horizontal line (purple) indicates the Centers for Medicare and Medicaid (CMS) DR screening target of 67%. *Separate chi-square tests of independence performed for January 2021 and June 2023 comparing WMPC to FHPC screening percentages. AI, artificial intelligence; DR, diabetic retinopathy.

DISCUSSION

This study examined the use of AI in DR screening in a rural primary care setting. Rural practices often operate in geographic isolation within a system of resource scarcity. Our clinic has one ophthalmologist located in the region.⁴ With the use of AI, our system was able to avoid > 80% of referrals to specialists for low-risk patients, ultimately reducing strain on specialty eye care.

Furthermore, our clinic has historically had poor rates of DR screening (40 – 60%). With access to AI DR screening, WMPC screening rates reached target level for the first time.

This paper adds to the growing body of research on the use of AI in DR screening and demonstrates feasibility in a rural primary care setting. After implementing AI DR screening, our screening rate exceeded the CMS target, and we achieved a high (74%) follow-up rate among those who screened positive. Future research should address the optimal referral timeline for high-risk patients and interventions to improve referral success, which is crucial for preventing vision loss. This cohort study was limited to a single healthcare system in one state with a modest sample size.

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Data Availability

Data available within the article or its supplementary materials.

References
